

Problem 1 **Laplace transform [10 pts]**



The Γ -function is defined through

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt, \quad x > 0.$$

a) Show that $\Gamma(1) = 1$ and $\Gamma(x+1) = x\Gamma(x)$, that is, $\Gamma(n+1) = n!$ for non-negative integers n .

b) Compute the Laplace transform of

$$f(x) = x^q$$

for any real number $q > -1$. Express this Laplace transform using the Γ -function.

c) The error function is given by

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt.$$

Show that the error function has Laplace transform

$$\mathcal{L}[\operatorname{erf}](s) = \frac{e^{s^2/4}(1 - \operatorname{erf}(s/2))}{s}.$$

Hint: Consider the Laplace transform of $\frac{d}{dx} \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} e^{-x^2}$ and use that $\operatorname{erf}(0) = 0$. Recall that $\int_0^\infty e^{-t^2} dt = \frac{\sqrt{\pi}}{2}$.

Grading aid

Solution

a) We find

$$\Gamma(1) = \int_0^\infty e^{-t} dt = \left[-e^{-t} \right]_0^\infty = 1$$

and

$$x\Gamma(x) = \int_0^\infty x t^{x-1} e^{-t} dt = \int_0^\infty \frac{d}{dt} t^x e^{-t} dt = \left[t^x e^{-t} \right]_0^\infty - \int_0^\infty t^x \frac{d}{dt} e^{-t} dt = \int_0^\infty t^x e^{-t} dt = \Gamma(x+1)$$

b) By direct computation we obtain

$$\int_0^\infty x^q e^{-sx} dx = \left/ sx = t \right/ = \int_0^\infty \left(\frac{t}{s}\right)^q e^{-t} \frac{1}{s} dt = \frac{1}{s^{q+1}} \int_0^\infty t^q e^{-t} dt = \frac{\Gamma(q+1)}{s^{q+1}}$$

c) We find

$$\begin{aligned} \mathcal{L} \left[\frac{d}{dx} \operatorname{erf}(x) \right] (s) &= \int_0^\infty \frac{2}{\sqrt{\pi}} e^{-x^2} e^{-sx} dx \\ &= \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-x^2 - sx - \frac{s^2}{4}} e^{\frac{s^2}{4}} dx \\ &= e^{\frac{s^2}{4}} \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-(x + \frac{s}{2})^2} dx \quad \left/ x + \frac{s}{2} = t \right/ \\ &= e^{\frac{s^2}{4}} \frac{2}{\sqrt{\pi}} \int_{s/2}^\infty e^{-t^2} dt \\ &= e^{\frac{s^2}{4}} \frac{2}{\sqrt{\pi}} \left(\int_0^\infty e^{-t^2} dt - \int_0^{s/2} e^{-t^2} dt \right) \\ &= e^{\frac{s^2}{4}} \left(1 - \operatorname{erf} \left(\frac{s}{2} \right) \right). \end{aligned}$$

By the properties of the Laplace transform we also know

$$\mathcal{L} \left[\frac{d}{dx} \operatorname{erf}(x) \right] (s) = s \mathcal{L} [\operatorname{erf}(x)] (s) - \operatorname{erf}(0) = s \mathcal{L} [\operatorname{erf}(x)] (s).$$

So,

$$\mathcal{L} [\operatorname{erf}(x)] (s) = \frac{e^{\frac{s^2}{4}} (1 - \operatorname{erf}(\frac{s}{2}))}{s}.$$

Grading manual

a) (2p)

b) (3p)

c) (5p)

△

Problem 2 Fourier series [10 pts]

Extend the function

$$f(x) = -x + 1,$$

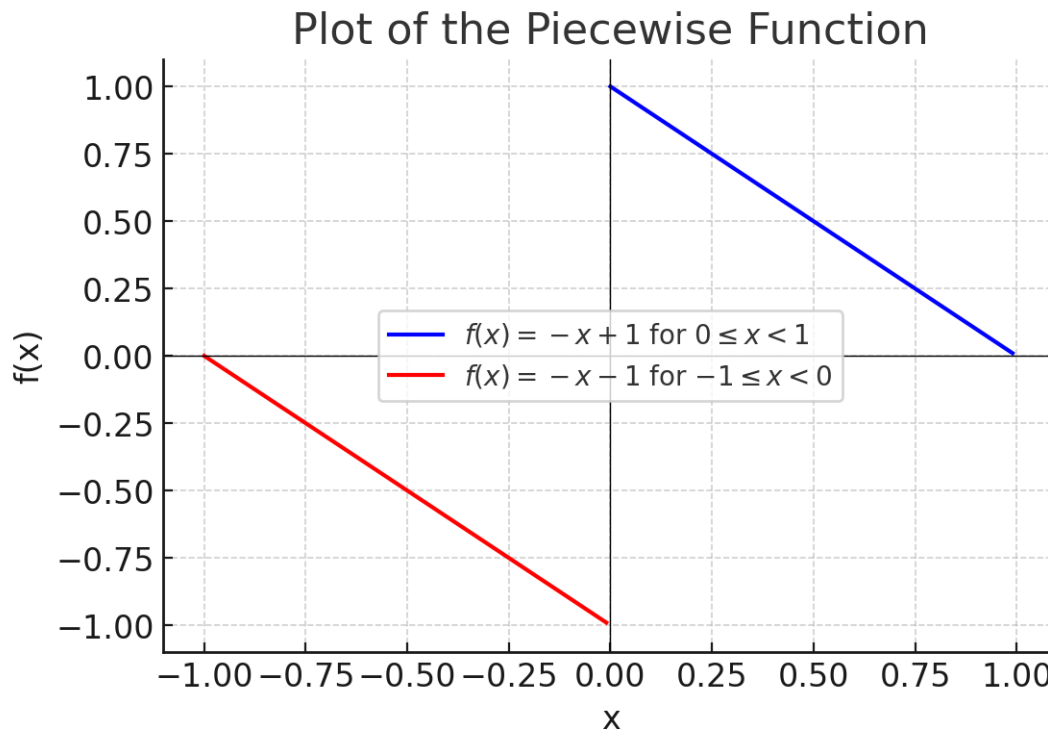
defined on the interval $[0, 1)$, to a periodic function in such a way that the extension has period 2 and is **odd**. Draw a plot of this new function on the interval $[-1, 1)$. Find the Fourier series of this new function, and explicitly write down the first 5 nonvanishing terms of the Fourier series.

Solution A function f is odd if $f(x) = -f(-x)$. Let $x \in [0, 1)$, then, according to its definition, $f(x) = -x + 1$. Because f is odd, $-f(x) = f(-x)$ and $-f(x) = x - 1 = f(-x)$. Let $y := -x$. Since $-x = y \in [-1, 0)$, we get $f(y) = -y - 1$.

We therefore get $f(-x) = -f(x)$. The resulting odd extension is

$$f(x) = \begin{cases} -x + 1 & 0 \leq x < 1 \\ -x - 1 & -1 \leq x < 0, \end{cases}$$

$f(x) = f(x + 2n)$ ($n \in \mathbb{Z}$), with period 2.



As the extension of f is odd, we have

$$a_n = 0, \quad n = 0, 1, 2, \dots$$

Moreover,

$$\begin{aligned} b_n &= \frac{1}{L} \int_{-L}^L \underbrace{f(x)}_{\text{odd}} \underbrace{\sin\left(\frac{\pi nx}{L}\right)}_{\text{odd}} dx \\ &= \frac{2}{L} \int_0^L f(x) \sin\left(\frac{\pi nx}{L}\right) dx \\ &= 2 \int_0^1 (-x+1) \sin(\pi nx) dx \\ &= -2 \int_0^1 x \sin(\pi nx) dx + 2 \int_0^1 \sin(\pi nx) dx \\ &= \underbrace{\left[2x \frac{1}{\pi n} \cos(\pi nx)\right]_0^1}_{=2/(\pi n) \cos(\pi n)} + \underbrace{\int_0^1 2 \frac{1}{\pi n} \cos(\pi nx) dx}_{=2/(\pi n)^2 \sin(\pi n)=0} + \underbrace{2 \int_0^1 \sin(\pi nx) dx}_{=-2/(\pi n) \cos(\pi n)+2/(\pi n)} \\ &= \frac{2}{\pi n} \end{aligned}$$

Therefore, we have for

$$\begin{aligned} n=1: \quad b_1 &= \frac{2}{\pi} \\ n=2: \quad b_2 &= \frac{2}{2\pi} = \frac{1}{\pi} \\ n=3: \quad b_3 &= \frac{2}{3\pi} \\ n=4: \quad b_4 &= \frac{2}{4\pi} = \frac{1}{2\pi} \\ n=5: \quad b_5 &= \frac{2}{5\pi}. \end{aligned}$$

The resulting Fourier series is

$$\begin{aligned} f(x) &= \sum_{n=1}^{\infty} \frac{2}{\pi n} \sin(n\pi x) \\ &\approx \frac{2}{\pi} \sin(\pi x) + \frac{1}{\pi} \sin(2\pi x) + \frac{2}{3\pi} \sin(3\pi x) + \frac{1}{2\pi} \sin(4\pi x) + \frac{2}{5\pi} \sin(5\pi x) + \dots \end{aligned}$$

Grading manual (2p) for the proper odd extension
(1p) for the plot
(3p) for b_n
(3p) for b_1, b_2, b_3, b_4, b_5
(1p) for the Fourier series.

Problem 3 Numerical methods for ODEs [10 pts]

- a) Verify that the function $f(x, y) = 2yx^{-4}$ satisfies a Lipschitz condition in the 2nd variable on the domain $1 \leq x < \infty$, $y \in \mathbb{R}$ and determine the associated Lipschitz constant.
- b) For the differential equation $y' = f(t, y)$, we consider the general two-stage *implicit* Runge–Kutta method

$$y_{n+1} = y_n + h \sum_{i=1}^2 b_i K_i,$$

with the stages

$$\begin{aligned} K_1 &= f(t_n + c_1 h, y_n + h a_{11} K_1 + h a_{12} K_2) \\ K_2 &= f(t_n + c_2 h, y_n + h a_{21} K_1 + h a_{22} K_2), \end{aligned}$$

corresponding to the Butcher tableau

$$\begin{array}{c|cc} c_1 & a_{11} & a_{12} \\ c_2 & a_{21} & a_{22} \\ \hline & b_1 & b_2 \end{array}$$

By applying this method to the scalar-valued differential equation

$$y' = \lambda y,$$

where λ is a constant, the K_1 and K_2 can be deduced as solution of the linear system

$$\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \begin{pmatrix} K_1 \\ K_2 \end{pmatrix} = \begin{pmatrix} \lambda y_n \\ \lambda y_n \end{pmatrix}.$$

- b1) Describe the matrix B with entries $\{b_{ij}\}_{1 \leq i, j \leq 2}$ explicitly, in terms of the $\{h a_{ij}\}_{1 \leq i, j \leq 2}$ and write down its determinant $\Delta := \det(B)$.
- b2) Assume that the determinant $\Delta \neq 0$. Find explicit formulas for K_1 and K_2 by solving the system

$$B \begin{pmatrix} K_1 \\ K_2 \end{pmatrix} = \begin{pmatrix} \lambda y_n \\ \lambda y_n \end{pmatrix}$$

Solution We see that

$$|f(x, y_1) - f(x, y_2)| = |2y_1x^{-4} - 2y_2x^{-4}| = 2x^{-4}|y_1 - y_2| \leq 2|y_1 - y_2|,$$

for all $x \in [1, \infty[$ and $y_1, y_2 \in \mathbb{R}$. The Lipschitz constant is therefore $L = 2$.

b1) For the differential equation $y' = \lambda y$, i.e., when $f(t, y) = \lambda y$, we find

$$\begin{aligned} K_1 &= \lambda(y_n + ha_{11}K_1 + ha_{12}K_2) \\ K_2 &= \lambda(y_n + ha_{21}K_1 + ha_{22}K_2). \end{aligned}$$

This implies the linear system

$$\begin{pmatrix} (1 - \lambda ha_{11})K_1 & -\lambda ha_{12}K_2 \\ -\lambda ha_{21}K_1 & (1 - \lambda ha_{22})K_2 \end{pmatrix} = \begin{pmatrix} \lambda y_n \\ \lambda y_n \end{pmatrix}. \quad (1)$$

It can be written in matrix form

$$(I_2 - \lambda hA) \begin{pmatrix} K_1 \\ K_2 \end{pmatrix} = \begin{pmatrix} \lambda y_n \\ \lambda y_n \end{pmatrix}, \quad (2)$$

which determines $B := (I_2 - \lambda hA)$, where I_2 is the 2×2 identity matrix and

$$A := \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

The determinant of $B = (I_2 - \lambda hA)$ computes

$$\Delta = 1 - \lambda h(a_{11} + a_{22}) + \lambda^2 h^2(a_{11}a_{22} + a_{12}a_{21}).$$

b2) We suppose that $\Delta \neq 0$. Then solving the 2×2 linear system (1) respectively (2) gives

$$K_1 = \frac{\lambda y_n(1 + \lambda h(a_{12} - a_{22}))}{\Delta} \quad K_2 = \frac{\lambda y_n(1 + \lambda h(a_{21} - a_{11}))}{\Delta}.$$

Grading manual

a) (2p)

b) (3p)

c) (5p)

Problem 4 Interpolation [10 pts]

Consider the points

$$\begin{array}{c|ccc} x_i & 0 & 1 & 2 \\ \hline y_i = f(x_i) & 1 & 2 & 0 \end{array}$$

Compute the cubic polynomial $p(x) = ax^3 + bx^2 + cx + d$ that satisfies the above interpolation conditions and the additional derivative condition $p'(1) = 3$.

Solution. We have $p'(x) = 3ax^2 + 2bx + c$. The system of equations is

$$\begin{aligned} a \cdot 0^3 + b \cdot 0^2 + c \cdot 0 + d &= d = 1 \\ a \cdot 1^3 + b \cdot 1^2 + c \cdot 1 + d &= a + b + c + d = 2 \\ a \cdot 2^3 + b \cdot 2^2 + c \cdot 2 + d &= 8a + 4b + 2c + d = 0 \\ 3a \cdot 1^2 + 2b \cdot 1 + c &= 3a + 2b + c = 3. \end{aligned}$$

From the first equation, we get directly that $d = 1$. Therefore:

$$\begin{aligned} a + b + c &= 1 \implies a = 1 - b - c \\ 8a + 4b + 2c &= -1 \implies 8(1 - b - c) + 4b + 2c = 8 - 4b - 6c = -1 \implies b = \frac{9}{4} - \frac{3}{2}c \\ 3a + 2b + c &= 3. \end{aligned}$$

Setting in the formulations of a and b in the third equation gives

$$\begin{aligned} 3(1 - b - c) + 2\left(\frac{9}{4} - \frac{3}{2}c\right) + c &= 3 - 3b - 3c + \frac{9}{2} - 3c + c = \frac{15}{2} - 3b - 5c = \frac{15}{2} - 3\left(\frac{9}{4} - \frac{3}{2}c\right) - 5c \\ &= \frac{15}{2} - \frac{27}{4} + \frac{9}{2}c - 5c = \frac{3}{4} - \frac{1}{2}c = 3 \implies c = -\frac{9}{2}. \end{aligned}$$

Then

$$b = \frac{9}{4} - \frac{3}{2}c = \frac{9}{4} - \frac{3}{2} \cdot \left(-\frac{9}{2}\right) = \frac{36}{4} = 9.$$

Finally,

$$a = 1 - b - c = 1 - 9 + \frac{9}{2} = -\frac{7}{2}.$$

Thus, the polynomial is given as

$$p(x) = -\frac{7}{2}x^3 + 9x^2 - \frac{9}{2}x + 1.$$

Grading manual 4 points for the system of equations, 1 point for the computation, 4 points for each unknown, 1 point for the polynomial.

Problem 5 Heat Equation [10 pts]

Consider the heat equation modeling a thin metal rod with isolated ends

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) = c^2 \frac{\partial^2}{\partial x^2} u(t, x), & \text{for } 0 \leq t, 0 \leq x \leq L, \\ \frac{\partial}{\partial x} u(t, 0) = \frac{\partial}{\partial x} u(t, L) = 0, & \text{for } 0 \leq t, \\ u(0, x) = f(x), & \text{for } 0 \leq x \leq L. \end{cases}$$

If $f(x) = \cos\left(\frac{n\pi}{L}x\right)$ then the solution is given by $u_n(t, x) = \cos\left(\frac{n\pi}{L}x\right) e^{-\left(\frac{n\pi c}{L}\right)^2 t}$ (n is a positive integer).

a) Show that the average temperature

$$\frac{1}{L} \int_0^L u_n(t, x) dx$$

is 0 for every t .

b) Show that the L^2 -norm (with respect to x)

$$\left(\int_0^L |u_n(t, x)|^2 dx \right)^{1/2}$$

is decreasing with t and find a T such that

$$\left(\int_0^L |u_n(T, x)|^2 dx \right)^{1/2} = \frac{1}{2} \left(\int_0^L |u_n(0, x)|^2 dx \right)^{1/2}.$$

Now consider a solution $u(t, x)$ for a general $f(x)$.

c) Show that the average temperature of $u(t, x)$ is constant with respect to t and that the L^2 -norm of $u(t, x)$ is non-increasing with t .

Grading aid

Solution

a) We directly compute

$$\frac{1}{L} \int_0^L u_n(t, x) dx = \frac{1}{L} \int_0^L \cos\left(\frac{n\pi}{L}x\right) e^{-\left(\frac{n\pi c}{L}\right)^2 t} dx = \frac{1}{L} e^{-\left(\frac{n\pi c}{L}\right)^2 t} \left[\frac{L}{n\pi} \sin\left(\frac{n\pi}{L}x\right) \right]_0^L = 0$$

b) For a general t we have

$$\int_0^L |u_n(t, x)|^2 dx = \int_0^L \cos^2\left(\frac{n\pi}{L}x\right) e^{-2\left(\frac{n\pi c}{L}\right)^2 t} dx = e^{-2\left(\frac{n\pi c}{L}\right)^2 t} \left[\frac{x + \frac{L}{2n\pi} \sin\left(\frac{2n\pi}{L}x\right)}{2} \right]_0^L = \frac{L}{2} e^{-2\left(\frac{n\pi c}{L}\right)^2 t}.$$

Thus we find the L^2 -norm

$$\left(\int_0^L |u_n(t, x)|^2 dx \right)^{1/2} = \sqrt{\frac{L}{2}} e^{-\left(\frac{n\pi c}{L}\right)^2 t}$$

and its derivative

$$\frac{d}{dt} \left(\int_0^L |u_n(t, x)|^2 dx \right)^{1/2} = \frac{\frac{d}{dt} \int_0^L |u_n(t, x)|^2 dx}{2 \left(\int_0^L |u_n(t, x)|^2 dx \right)^{1/2}} = - \left(\frac{n\pi c}{L} \right)^2 \sqrt{\frac{L}{2}} e^{-\left(\frac{n\pi c}{L}\right)^2 t} < 0,$$

so the L^2 -norm is decreasing. We also want to find T such that

$$e^{-\left(\frac{n\pi c}{L}\right)^2 T} = \frac{1}{2},$$

that is,

$$T = \frac{\ln\left(\frac{1}{2}\right)}{-\left(\frac{n\pi c}{L}\right)^2} = \frac{L^2 \ln(2)}{(n\pi c)^2}.$$

c) We use the equation and boundary conditions to obtain

$$\frac{\partial}{\partial t} \frac{1}{L} \int_0^L u(t, x) dx = \frac{1}{L} \int_0^L \frac{\partial}{\partial t} u(t, x) dx = \frac{c^2}{L} \int_0^L \frac{\partial^2}{\partial x^2} u(t, x) dx = \frac{c^2}{L} \left[\frac{\partial}{\partial x} u(t, x) \right]_0^L = 0$$

and

$$\begin{aligned} \frac{\partial}{\partial t} \frac{1}{L} \int_0^L |u(t, x)|^2 dx &= \frac{2}{L} \int_0^L u(t, x) \frac{\partial}{\partial t} u(t, x) dx \\ &= \frac{2c^2}{L} \int_0^L u(t, x) \frac{\partial^2}{\partial x^2} u(t, x) dx \\ &= \frac{2c^2}{L} \left[u(t, x) \frac{\partial}{\partial x} u(t, x) \right]_0^L - \frac{2c^2}{L} \int_0^L \frac{\partial}{\partial x} u(t, x) \frac{\partial}{\partial x} u(t, x) dx \\ &= -\frac{2c^2}{L} \int_0^L \left| \frac{\partial}{\partial x} u(t, x) \right|^2 dx \leq 0 \end{aligned}$$

Alternatively the last part can be solved using the formula for the general solution $u(x, t) = \hat{f}_0 + \sum_{n=1}^{\infty} \hat{f}_n u_n(t, x)$, where $\hat{f}_i$ are the Fourier coefficients of f using the facts about $u_n(t, x)$ from the first two parts and the orthogonality of the $u_n(t, x)$.

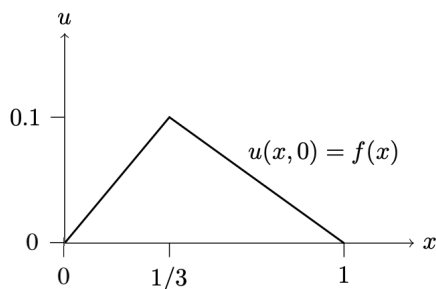
Grading manual

- a) (2p)
- b) (3p) - (2p) for showing that the L^2 norm is decreasing and (1p) for finding T
- c) (5p) - (2p) for showing that the average temperature is constant and (3p) for showing that the L^2 norm is decreasing.

△

Problem 6 Wave Equation [10 pts]

We consider a string of length $L = 1$ with its ends fixed at $x = 0$ and $x = 1$. The string is set to vibrate from rest by releasing it from an initial triangular shape $f(x)$:



- a) Compute the coefficients b_n and b_n^* in the solution

$$u(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{L}x\right) \left(b_n \cos(\lambda_n t) + b_n^* \sin(\lambda_n t) \right)$$

to the wave equation

$$u_{tt} = \frac{1}{\pi^2} u_{xx}.$$

- b) Determine the λ_n .
c) Write down the final expression for $u(x, t)$.

Solution

- a) The initial shape function is

$$f(x) = \begin{cases} \frac{3}{10}x, & 0 \leq x \leq \frac{1}{3} \\ \frac{3(1-x)}{20}, & \frac{1}{3} \leq x \leq 1. \end{cases}$$

This corresponds to $u(x, 0) = f(x)$ for $0 < x < L$. From this one computes

$$\begin{aligned} b_n &= 2 \int_0^1 f(v) \sin(n\pi v) dv \\ &= \frac{3}{5} \int_0^{\frac{1}{3}} v \sin(n\pi v) dv + \frac{3}{10} \int_{\frac{1}{3}}^1 (1-v) \sin(n\pi v) dv \\ &= -\frac{\cos(\frac{n\pi}{3})}{5n\pi} + \frac{3 \sin(\frac{n\pi}{3})}{5 n^2 \pi^2} + \frac{\cos(\frac{n\pi}{3})}{5n\pi} + \frac{3 \sin(\frac{n\pi}{3})}{10 n^2 \pi^2} \\ &= \frac{9}{10\pi^2} \frac{\sin(\frac{n\pi}{3})}{n^2}. \end{aligned}$$

The string starts from rest, i.e., $u_t(x, 0) = g(x) = 0$ for $0 < x < L$, therefore the coefficients $b_n^* = 0$.

b) Recall the general expression

$$\lambda_n = c \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots$$

In the the problem, we assume $c = 1/\pi$ (recall the general wave equation $u_{tt} = c^2 u_{xx}$) and $L = 1$, which implies that $\lambda_n = n$.

c) The complete solution is

$$u(x, t) = \frac{9}{10\pi^2} \sum_{n>0} \frac{\sin(\frac{n\pi}{3})}{n^2} \sin n\pi x \cos nt.$$

Grading manual (10p)

a) (6p)

b) (2p)

c) (2p):

Problem 7 Fourier Transform [10 pts]

a) Show for $a \neq 0$ that

$$\mathcal{F}(f(at))(\omega) = \frac{1}{|a|} \mathcal{F}(f(t))\left(\frac{\omega}{a}\right).$$

b) Consider the function

$$\tilde{h}(\omega) = \frac{\exp(-\omega^2)}{1 + \omega^2}.$$

Write the given $\tilde{h}$ as a product of two familiar Fourier transforms $\tilde{f}$ and $\tilde{g}$. Identify the functions f and g and express the function h as a convolution.

Solution

a) We must consider the cases $a > 0$ and $a < 0$.

$a > 0$:

$$\begin{aligned} \mathcal{F}(f(at))(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(ax) \exp(-i\omega x) dx \\ &= \frac{1}{a\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \exp(-i\frac{\omega}{a}x) dx \\ &= \frac{1}{a} \mathcal{F}(f(t))\left(\frac{\omega}{a}\right) \end{aligned}$$

Here we used a change of variables: $ax = y$, $adx = dy$.

$a < 0$:

$$\begin{aligned} \mathcal{F}(f(at))(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(ax) \exp(-i\omega x) dx \\ &= \frac{1}{a\sqrt{2\pi}} \int_{\infty}^{-\infty} f(x) \exp(-i\frac{\omega}{a}x) dx \\ &= -\frac{1}{a} \mathcal{F}(f(t))\left(\frac{\omega}{a}\right) \end{aligned}$$

From this we conclude that

$$\mathcal{F}(f(at))(\omega) = \frac{1}{|a|} \mathcal{F}(f(t))\left(\frac{\omega}{a}\right).$$

b)

$$\tilde{h} = \tilde{f}\tilde{g} = \exp(-\omega^2) \frac{1}{1 + \omega^2}.$$

$$\tilde{f} = \mathcal{F}\left(\frac{1}{\sqrt{2}} \exp\left(-\frac{t^2}{4}\right)\right)$$

$$\tilde{g} = \mathcal{F}\left(\sqrt{\frac{\pi}{2}} \exp(-|t|)\right)$$

Recall that we defined the convolution product without the $\frac{1}{\sqrt{2\pi}}$ coefficient. Therefore

$$\begin{aligned} h(t) &= f * g(t) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2}} \exp\left(-\frac{(t-x)^2}{4}\right) \sqrt{\frac{\pi}{2}} \exp(-|x|) dx \\ &= \frac{1}{2\sqrt{2}} \int_{-\infty}^{\infty} \exp\left(-\frac{(t-x)^2}{4}\right) \exp(-|x|) dx \end{aligned}$$

Grading manual (10p)

a) (6p)

b) (4p)

Problem 8 Numerical integration [10 pts]

We consider a quadrature formula on the interval $[-1, 1]$ which uses the quadrature points $x_0 = -a$ and $x_1 = a$, where $0 < a \leq 1$:

$$\int_{-1}^1 f(s)ds \approx w_0 f(-a) + w_1 f(a).$$

- a) The formula is required to be exact whenever f is a polynomial of degree 1. Show that $w_0 = w_1 = 1$, independent of the value of a .
- b) Find the particular value of a for which the formula is exact also for all polynomials of degree 2.
- c) Show that for the particular value of a for which the formula is exact for all polynomials of degree 2, the formula is also exact for all polynomials of degree 3.

Solution Recall that a quadrature formula is exact for all polynomials of degree n iff, it is exact for the particular functions $f(x) = x^r$, $r = 0, \dots, n$. Hence, we must have

$$\int_{-1}^1 s^r ds = w_0 (-a)^r + w_1 a^r,$$

from which we deduce

$$\frac{1 - (-1)^{r+1}}{r+1} = a^r ((-1)^r w_0 + w_1).$$

- a) From the last identity we deduce for a polynomial of degree 1 the system

$$\begin{aligned} r = 0 : \quad & 2 = w_0 + w_1 \\ r = 1 : \quad & 0 = a(-w_0 + w_1). \end{aligned}$$

Since $a > 0$, this implies $w_0 = w_1 = 1$.

- b) With respect to polynomials of degree 2, the above system is extended by another relation

$$r = 2 : \quad \frac{2}{3} = 2a^2,$$

which implies $a = \frac{1}{\sqrt{3}}$.

- c) Going to polynomials of degree 3, we have one more equation

$$r = 3 : \quad 0 = 2a^3(-1 + 1),$$

which is true for the particular value of $a = \frac{1}{\sqrt{3}}$. Hence the formula is exact for x^r , $r = 0, 1, 2, 3$, i.e., it is exact for every polynomial of degree 3 iff $a = \frac{1}{\sqrt{3}}$.

Grading manual (10p)

a) (4p)

b) (3p)

c) (3p)

Problem 9 Numerics for Nonlinear Equations [10 pts]

Let the iteration function $g(x) : (0, \infty) \rightarrow \mathbb{R}$ be given by

$$g(x) = \frac{x}{2} + \frac{1}{x}.$$

Determine the largest interval on which g fulfills the contraction property $|g'(x)| \leq L < 1$ for some $L > 0$. Compute the fixed point in this interval. Is the sequence $(x_k)_{k \in \mathbb{N}}$ with $x_k = g(x_{k-1})$, $k \in \mathbb{N}$ quadratically convergent? Explain your reasoning.

Solution We compute the derivative $g'(x) = \frac{1}{2} - \frac{1}{x^2}$ and for the fulfillment of the contraction property, we must have

$$|g'(x)| \leq L < 1,$$

which implies

$$-1 < \frac{1}{2} - \frac{1}{x^2} < 1,$$

which is equivalent to

$$x > \sqrt{\frac{2}{3}} =: q.$$

Furthermore, the derivative is strictly increasing and bounded above by $\frac{1}{2}$. Thus, g is a contracting self-mapping on any closed interval $[a, \infty)$ with $a > q$ and the sequence x_k therefore converges to the uniquely determined fixed point, which is easily computed as $\tilde{x} = \sqrt{2}$:

$$\tilde{x} = \frac{\tilde{x}}{2} + \frac{1}{\tilde{x}} \implies \frac{\tilde{x}}{2} = \frac{1}{\tilde{x}} \implies \frac{\tilde{x}^2}{2} = 1 \implies \tilde{x}^2 = 2,$$

and hence $\tilde{x} = \sqrt{2}$ ($\tilde{x} = -\sqrt{2}$ is excluded because the function g is defined on $(0, \infty)$).

To verify quadratic convergence, we compute

$$\frac{|x_{k+1} - \tilde{x}|}{|x_k - \tilde{x}|^2} = \frac{|\frac{x_k}{2} + \frac{1}{x_k} - \sqrt{2}|}{x_k^2 - 2\sqrt{2}x_k + 2} = \frac{|\frac{x_k}{2} + \frac{1}{x_k} - \sqrt{2}|}{2x_k \underbrace{\left(\frac{x_k}{2} - \sqrt{2} + \frac{1}{x_k}\right)}_{\geq 0}} = \frac{1}{2x_k} \leq \frac{1}{2q},$$

we can use q as the lower bound for x_k . This confirms that the sequence exhibits quadratic convergence.

Grading manual 1 point for the derivative of g , 3 points for the contraction property, 3 points for the fixed point, 3 points for quadratic convergence.

Grading Scale

A	48–90
B	42–47
C	35–41
D	29–34
E	22–28
F	0–21